sutd404

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1 Introduction

The Singapore University of Technology and Design (SUTD) has over 200 rooms split across 5 buildings. To help people (especially visitors) find rooms, each room is assigned an address of the form

```
x.y-z
```

where `x` is a building number, `y` is a floor number, and `z` is a room number. However, resources that aim to map rooms to addresses remain few and far between. Physical resources like campus signs exist, but require one to walk around, expending lots of energy and time. Digital resources like the SUTD Assistive Maps (SAM) Telegram bot exist, but often fail on noisy input, which can be especially common for visitors unfamiliar with typing in English or the SUTD rooms themselves. To fill this gap, we built SUTD 404, a digital resource that aims to map rooms to addresses under noise.

For more information, visit **github.com/gregorylimeurhen/sutd404** for our code repository, and **sutd404.vercel.app** for our web application.

2 Methodology

Let V be a vocabulary, and $R \subseteq V^*$ be a room set totally ordered by $\prec$, and $A \subseteq V^*$ be an address set, and $B \subseteq V$ be a boundary set, and $C \in \mathbb{N}$ be a corruption count, and $\alpha \in \mathbb{R}_{[0,1]}$ be a transposition rate, and $\beta \in \mathbb{R}_{[0,1]}$ be a substitution rate, and $\boldsymbol{\pi} = (\pi_{\text{train}}, \pi_{\text{val}}, \pi_{\text{test}}) \in \Delta^2$ be a training-validation-testing split, and $I : V \rightharpoonup \mathbb{Z}^2$ be a keyboard-position map, and $C\boldsymbol{\pi} \in \mathbb{N}^3$, and $N \in \mathbb{N}_{\geq}$ be a result count.

Definition 1 (Keyboard)**.**

$$K = \begin{bmatrix} \texttt{`} & \texttt{1} & \texttt{2} & \texttt{3} & \texttt{4} & \texttt{5} & \texttt{6} & \texttt{7} & \texttt{8} & \texttt{9} & \texttt{0} & \texttt{-} & \texttt{=} \\ \texttt{q} & \texttt{w} & \texttt{e} & \texttt{r} & \texttt{t} & \texttt{y} & \texttt{u} & \texttt{i} & \texttt{o} & \texttt{p} & \texttt{[} & \texttt{]} & \texttt{\backslash} \\ & \texttt{a} & \texttt{s} & \texttt{d} & \texttt{f} & \texttt{g} & \texttt{h} & \texttt{j} & \texttt{k} & \texttt{l} & \texttt{;} & \texttt{'} & \\ & \texttt{z} & \texttt{x} & \texttt{c} & \texttt{v} & \texttt{b} & \texttt{n} & \texttt{m} & \texttt{,} & \texttt{.} & \texttt{/} & & \\ & & & & & \texttt{␣} & \texttt{␣} & \texttt{␣} & \texttt{␣} & \texttt{␣} & & & \end{bmatrix}$$

Our keyboard uses the 60% ANSI QWERTY en-US keyboard layout without backspace, tab, caps lock, enter, shift, ctrl, alt, and system keys (Wikipedia contributors, 2026). All keys have equal padding between them.

Algorithm 1 CORRUPT($\boldsymbol{x}$)

Require: $\boldsymbol{x} \in V^*$
$(T_L, T_M, T_R) \leftarrow (V^*B \cup \{()\}, (V \setminus B)^+, BV^* \cup \{()\})$
$T \leftarrow \{\boldsymbol{uwbvz} \mid \boldsymbol{u} \in T_L,\ \boldsymbol{v}, \boldsymbol{w} \in T_M,\ \boldsymbol{z} \in T_R,\ b \in B,\ \boldsymbol{x} = \boldsymbol{uvbwz}\} \setminus \{\boldsymbol{x}\}$
if $T \neq \varnothing$ **then** $\boldsymbol{x} \sim \alpha\,\mathcal{U}(T) + (1-\alpha)\,\delta(\boldsymbol{x})$
for $i \in [1..|\boldsymbol{x}|]$ **do**
  $S \leftarrow \{k \in K \setminus \{x_i\} \mid I(k) - I(x_i) \in [-1..1] \times [-2..2]\}$
  if $S \neq \varnothing$ **then** $x_i \sim \beta\,\mathcal{U}(S) + (1-\beta)\,\delta(x_i)$
return $\boldsymbol{x}$

Algorithm 2 SPLIT()

$(D_{\text{train}}, D_{\text{val}}, D_{\text{test}}) \leftarrow (\varnothing, \varnothing, \varnothing)$
for $r \in R$ **do**
  $P \leftarrow \varnothing$
  while $|P| < C$ **do**
    $r' \sim$ CORRUPT(r)
    if $r' \neq r$ **then** $P \leftarrow P \cup \{(r', r)\}$
  $P_{\text{train}} \sim \mathcal{U}(\{Q \subseteq P \mid |Q| = \pi_{\text{train}} C\})$
  $P_{\text{val}} \sim \mathcal{U}(\{Q \subseteq P \setminus P_{\text{train}} \mid |Q| = \pi_{\text{val}} C\})$
  $P_{\text{test}} \leftarrow P \setminus (P_{\text{train}} \cup P_{\text{val}})$
  $(D_{\text{train}}, D_{\text{val}}, D_{\text{test}}) \leftarrow (D_{\text{train}} \cup P_{\text{train}}, D_{\text{val}} \cup P_{\text{val}}, D_{\text{test}} \cup P_{\text{test}})$
return $(D_{\text{train}}, D_{\text{val}}, D_{\text{test}})$

Our corruption algorithm corrupts strings by transposing spans and substituting characters. First, the algorithm samples whether to transpose spans by uniformly sampling from the space of span transpositions. Then, for each character in the string, the algorithm samples whether to substitute the character by uniformly sampling from the space of character substitutions. This space contains nearby characters collected by sliding a window of height 3 and width 5 centred at each position in K. Finally, the algorithm returns the string. We set $C = 50$, and $\alpha = 0.5$, and $\beta = 0.05$, and $\boldsymbol{\pi} = (0.8, 0.1, 0.1)$ to generate plausible-looking typos in a tractable way.

Definition 2 (Denoiser)**.** Let $f : V^* \times R \to \mathbb{R}$ be a scorer.

$$\mathscr{D}_f : V^* \to R$$
$$\mathscr{D}_f(x) = \arg\max_{r \in R} f(x, r)$$

Our denoiser is a Nanochat-parameterised, character-level transformer of depth 2 with decoding constrained to the trie induced by R (Karpathy, 2025). We use character-level tokenisation as typos are often character-level, and room names are short enough for individual characters to be highly meaningful semantic units for our task. We use trie-constrained decoding to efficiently avoid hallucination caused by unconstrained decoding. To train our denoiser, we use a standard cross-entropy loss and AdamW optimiser on default settings. A more complete list of hyperparameters can be found in our repository.

Definition 3 (Lookup)**.**

$$\mathscr{L} : R \to A$$

Our lookup is a standard room-address table.

Algorithm 3 $\textsc{Infer}(f, x)$

Require: $f : V^* \times R \to \mathbb{R}$, $x \in V^*$
  if $x \notin R$ **then** $x \leftarrow \mathscr{D}_f(x)$
  $x \leftarrow \mathscr{L}(x)$
  return x

Our inference algorithm receives a scorer and string as input. Firstly, the algorithm denoises the string to a room using the scorer if the string is not a room. Then, the algorithm looks up the address of the room. Finally, the algorithm returns the address.

We first evaluated our denoiser against identity (I), substring Jaccard similarity (SJ), negative Damerau–Levenshtein distance (DL), and longest common prefix length (LCP) denoisers, measuring the accuracy and mean latency of each denoiser on the test set. To maximise fairness and reproducibility, we fixed a seed and hardware.

Denoiser	**Accuracy** (%)	**Mean Latency** (ms)
LCP	29.2	**0.04**
DL	71.9	16.7
SJ	89.3	20.8
Ours	**97.3**	8.19

Table 1: Denoiser evaluation results summary.

Accuracy was measured using the fraction of test inputs for which final address equals gold address. Mean latency was measured using the mean time taken to get an output from giving an input.

We implemented our system as a web application. We set $N = 26$ because, grouping rooms by shared prefix up to terminal numbers, the largest group (the

group of Think Tank rooms) contains 26 rooms, so showing 26 results could allow a user to find their room by prefix without disambiguating by terminal numbers.

Algorithm 4 PREDICT(x)

Require: $x \in V^*$
  $x \leftarrow \textsc{Strip}(x)$
  $x \leftarrow \textsc{Depunctuate}(x)$
  $x \leftarrow \textsc{Lowercase}(x)$
  $x \leftarrow \textsc{AlphanumericSplit}(x)$
  $x \leftarrow \textsc{Alias}(x)$
  if $x \in R$ **then return** $((x, \mathscr{L}(x)))$
  $\boldsymbol{f} \leftarrow (\mathrm{LCP}, \mathrm{SJ}, \mathrm{DL}, \textsc{Ours})$
  for $r \in R$ **do**
    for $f \in \boldsymbol{f}$ **do**
      $p(r, f) \leftarrow |\{r' \in R \mid f(x, r') > f(x, r)\}| + 1$
  for $r \in R$ **do**
    $s(r) \leftarrow \sum_{i=1}^{|\boldsymbol{f}|} (1/|R|^{i-1})(|R| - p(r, f_i))$
  $\boldsymbol{o} \leftarrow \min \arg\max_{\boldsymbol{r} \in R^k_{\preceq}} \sum_{i=1}^{3} s(r_i)$
  return $((o, \mathscr{L}(o)) \mid o \in \boldsymbol{o})$

Our prediction algorithm receives a string as input. Firstly, the algorithm normalises the string by stripping whitespace, removing punctuation, lowercasing, separating letters from numbers, and replacing words with their aliases, to make the string more closely match training strings. Then, the algorithm computes top-ranked rooms. Finally, the algorithm looks up the address of each room, returning top-ranked room-address pairs.

To more deeply improve user experience, we also implement the following features to make the application more efficient, responsive, and robust.

- **Incremental prediction.** We predict on each new character entered.
- **Prefix caching.** We KV-cache.

We further evaluated our system against SAM, measuring accuracy on a held-out set obtained by 10 inputs from our test set using the Python library `random` with UTF-8 encoding. To maximise fairness and reproducibility, we fixed the same seed and hardware as before.

Input	design bdhavioural laboratory 1
Gold	behavioural design laboratory 1
Prediction	–
Input	capstone rolm 8
Gold	capstone room 8
Prediction	capstone room 8, capstone room 7, capstone room 6, capstone room 5
Input	lecturr 5 theatre
Gold	lecture theatre 5
Prediction	lecture theatre 5, lecture theatre 4, lecture theatre 2, lecture theatre 3
Input	com;ositemdesign laboratory
Gold	composite design laboratory
Prediction	–
Input	thinl tank 23
Gold	think tank 23
Prediction	think tank 15, think tank 24, a c toh think tank 25, think tank 22
Input	temasekvlaboratories and itrust labiratkry 6
Gold	temasek laboratories and itrust laboratory 6
Prediction	temasek laboratories and itrust laboratory 6
Input	think 2mtank
Gold	think tank 2
Prediction	think tank 2, think tank 14
Input	section compos7te
Gold	composite section
Prediction	cohort classroom 7
Input	hokkien lecture coundayioh theatre 4
Gold	hokkien foundation lecture theatre 4
Prediction	hokkien foundation lecture theatre 4, lecture theatre 2, lecture theatre 3, lecture theatre 1
Input	cohference capsrone ro9m 2
Gold	capstone conference room 2
Prediction	capstone room 9

Table 2: SAM evaluation results.

The results suggest that SAM, as a traditional matching-based algorithm, can be effective on mildly corrupted inputs with strong lexical overlap—occasionally recovering the exact target—but degrades markedly as noise becomes more se-

vere. We conclude that SAM can be robust to limited orthographic variation, but can also fail to generalise beyond surface-form similarity.

System	**Accuracy** (%)
SAM	40
Ours	**100**

Table 3: System evaluation results summary.

Our system outperformed SAM, achieving complete recovery. We believe these results demonstrate strong promise for the task of mapping rooms to addresses under noise, and provide a practical foundation for reliable campus assistance.

3 Appendix

Notation 1. Let $n \in \mathbb{N}_{\geq 1}$.

$$R^n_{\prec} = \{\boldsymbol{r} \in R^n \mid r_1 \prec \cdots \prec r_n\}$$

References

Andrej Karpathy. nanochat: The best chatgpt that $100 can buy. `https://github.com/karpathy/nanochat`, 2025.

Wikipedia contributors. Qwerty. `https://en.wikipedia.org/wiki/QWERTY`, 2026.